Existing System

Cloud resources are managed “manually”.

Uses “basic monitoring dashboards”.

Fixed rule-based resource allocation.

No prediction of future usage.

#Results in “high cloud cost”.

Disadvantages of Existing System ❌

- ⛔ Manual and time-consuming resource management
- ⛔ Limited monitoring with basic dashboards
- ⛔ Inflexible rule-based allocation
- ⛔ No intelligent usage prediction
- ⛔ Increased resource wastage and high cost

Proposed System

✅Uses “Machine Learning techniques”

1. Cloud Usage Prediction 📊

- The system analyzes historical cloud usage data (CPU, memory, traffic) using machine learning to forecast future resource requirements.

Example: The system predicts that next week’s workload will need 50% more CPU and memory.

2. Idle and Underutilized Resource Detection 🔍

- The system minimizes cloud cost through idle resource detection, rightsizing, and ML-based predictive scaling.

Example: Before optimization: ₹10,000/month
After optimization: ₹6,200/month
Savings achieved: 38%

3. Cloud Cost Optimization 💰

- The system automatically adjusts cloud resources based on predictions and usage analysis to reduce overall cost.

Example: Idle virtual machines are shut down or resized to lower-cost instances.